Peter Lubell-Doughtie

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18+ years of technical expertise leading fast-paced initiatives, managing high-growth teams, and delivering AI integrations. Skilled in programming with languages such as Java, Python, and Clojure. Led a team of over 40 engineers in platform building and implementing ML AI solutions. Increased revenue at Ona by over 40% per year from 2017 to 2021.

WORK EXPERIENCE

Ona **November 2013 - Present**
Co-Founder and CTO **New York, New York USA**

Increased revenue by over 40% per year from 2017-2021.

- Hands-on leadership in technology strategy, innovation, recruitment, leveling, and mentorship to grow our high-performing engineering org to over 40 people.
- Led system design and product development, implementing best practices to create data and health platforms, storing over 500,000 patients' secure data within 12 months.
- Integrated ML AI solutions, including unsupervised health risk scoring, computer vision for medical test results, generative AI transcription, and medical history summarization.
- Led collaboration and communication in partnership with key stakeholders, defined accountable business goals, negotiated contracts, and led financial management.
- Programmed in Clojure, ClojureScript, CQL, FHIR, Java, Kotlin, Python. Scalable system architecture and API design. Tracked errors, quality assurance metrics.

Mira Financial **January 2018 - April 2019**
Advisor **New York, New York USA**

- Advised internal team on data engineering, AI, scalability, and health systems needs. Communicated technical design to external partners.
- Interviewed candidates, mentored the technical team, collaborated on system design.

Intent Media **November 2012 - November 2013**
Lead Data Engineer **New York, New York USA**

- Led the data engineering team in building a data platform for ETL, analysis, and ML at scale. Managing prioritization between model and pipeline building.
- Communicated the data team's roles, needs, and accomplishments to the executive team and the company.
- Researched and published a novel distributed logistic regression implementation based on BFGS.
- Programmed ML models, data pipelines, and platform integrations in Clojure, Cascalog, Java, Pig, and Ruby.

Columbia University, Modi Research Group **December 2011 - October 2013**
Senior Software Engineer **New York, New York USA**

- Built semi-automated data analysis and management tools for energy infrastructure and development planning in Python and R.
- Developed web applications and web services in Python using Django for mobile data collection and management.
- Project managed remote teams and coordinated with corporate, non-profit, and government project partners.

Princeton University **October 2011 - April 2012**
Machine Learning Researcher **Princeton, New Jersey USA**

- Member of the CITP, reported to Edward Felton and Matthew Salganik.
- Collaborated with Nadia Heninger and other computer science graduate students to create an online sociology laboratory.
- Developed a REST and XML based web service, as well as Ruby on Rails and JavaScript web applications and plugins.
- Data science and data analysis using Ruby, Python, and R. Wrote algorithms for data processing in Ruby and C.
- Researched and implemented pair-wise algorithms for processing large amounts of vote data to determine user preferences.

Helioid **April 2008 - December 2011**
Co-Founder **New York, New York USA**

- Developed and integrated internet-facing learning algorithms in Ruby, as well as web applications and APIs in Ruby on Rails.
- Responsible for business development and administration, spoke at conferences and published papers to promote the company and its research.
- Researched and implemented technologies in information retrieval, text analysis, statistics, and machine learning.

Mobile Input Devices and Systems **April 2008 - June 2009**
Managing Engineer **Alameda, California USA**

- Participated in strategic planning and business development.
- Managed software development of chorded typing glove and typing pad, written in C and Assembly.
- Implemented and maintained website in Ruby on Rails and JavaScript.

Self Employed **December 2007 - March 2008**
Consultant **New York, New York USA**

- Maintained web applications in Ruby on Rails and advised on requirements for secure web applications.
- Wrote programmer evaluation tests for Ruby, Ruby on Rails, and JavaScript.
- Recommended additions and modifications to existing internet products and services.
- Developed and implemented websites and content management systems in PHP, Ruby, and SQL.
- Edited graphics and layouts using Adobe Photoshop and Adobe Illustrator.

Netsprout **June 2006 - November 2007**
Lead Developer **San Francisco, California USA**

- Responsibilities included team management, client interaction, individual and collaborative development.
- Programmed web applications, APIs, and RSS feeds in Ruby on Rails, AJAX, Actionscript, PHP, and JSP; desktop application in Java for organ transplant matching.
- Site reliability engineering, including load balancing system design, performance testing, database management, static code analysis.
- Developed Ruby on Rails plug-ins, transcoding systems using FFMPEG, XML APIs, RSS feed implementation.

Stanford University **June 2006 - December 2006**
Research Assistant **Stanford, California USA**

- Programming and data science in Ruby and Python to organize academic research into databases, flow charts, and text summaries.

EDUCATION

MSc, Artificial Intelligence, Cum Laude
University of Amsterdam — Amsterdam, Netherlands • January 2011

Track : Learning Systems, Thesis : Learning to Rank from Relevance Feedback

BS, Symbolic Systems
Stanford University — Stanford, California USA • January 2006

Concentration : Decision Making and Rationality

PUBLICATIONS

A full-STAC remedy for global digital health transformation : open standards, technologies, architectures, and content — December 2023
Oxford Open Digital Health

Low frequency of Plasmodium falciparum hrp2/3 deletions from symptomatic infections at a primary healthcare facility in Kilifi, Kenya. Frontiers in Epidemiology — February 2023
Frontiers in Epidemiology

Digitalization of routine health information systems : Bangladesh, Indonesia, Pakistan — August 2022
Bulletin of the World Health Organisation

Transforming Rapid Diagnostic Tests for Precision Public Health : Open Guidelines for Manufacturers and Users — July 2022
JMIR Biomed Eng

The Design and Evaluation of a Mobile System for Rapid Diagnostic Test Interpretation — March 2021
Proceedings of the ACM on Interactive Mobile Wearable and Ubiquitous Technologies

Practical Distributed Classification using the Alternating Direction Method of Multipliers Algorithm — October 2013
IEEE BigData

Improving Data Collection and Monitoring through Real-time Data Analysis — January 2013
ACM DEV

Learning to Rank from Relevance Feedback for e-Discovery — April 2012
ECIR 2012 - 34th European Conference on Information Retrieval

Improving Result Diversity using Probabilistic Latent Semantic Analysis — May 2011
DIR 2011 - Dutch-Belgian Information Retrieval Workshop

Helioid at TREC-Legal 2011 : Learning to Rank from Relevance Feedback for e-Discovery — January 2011
Proceedings of the 2011 Text REtrieval Conference

Using Genetic Programming to Evolve a General Purpose Sorting Network for Comparable Data Sets — June 2003
Genetic Algorithms and Genetic Programming at Stanford 2003

SKILLS

Programming Languages: AWS, Beam, C, C++, Clojure (Script), CQL, GCP, Hetzner, Java, JavaScript, Kotlin, Lisp, Perl, PHP, Pig, Python, R, Ruby, SH, SQL, TypeScript

AI/ML: Keras, langchain, Matlab, Mediapipe, Pandas, SciPy

Software: Ansible, Apache, Cascalog, Django, Docker, FFMPEG, Hadoop, HAPI FHIR, Helm, Kubernetes, Nginx, NiFi, React, Ruby on Rails, Spark, Terraform

Datastores: Kafka, MySQL, Parquet, PostgreSQL, Sqlite

Regulatory: GDPR, HIPAA

Management: Agile, Coaching, Mentorship, Performance reviews, Recruitment, Resourcing, Retention, Roadmaps